NexusOS

AI Business Operating System

Part 03.0

Linear Issue WAL-8

Requirements Analysis

Project	NexusOS — AI Business Operating System
Part	Part 03.0
Linear issue	WAL-8
Report title	Requirements Analysis
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Purpose	Standalone Linear attachment (manual upload)
Source report(s)	docs/reports/Part-03.0/Part-03.0-Requirements-Analysis.md

This PDF is a presentation export of repository engineering report content. Source report files in the repository were not modified.

Evidence Statement

This report is a retrospective engineering reconstruction produced after implementation. Its contents are derived exclusively from repository evidence. It is not a contemporaneous engineering document. No engineering events are fabricated. Where repository evidence is unavailable, that limitation is explicitly documented.

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Report Content

Content below is copied from the source repository report(s) without alteration.

Part 03.0 — Requirements Analysis

Linear: WAL-8 **Reconstruction date:** 2026-07-19

Evidence Statement

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Objective

Define the requirements for an enterprise multi-tenant database architecture before modeling and implementing tables.

Requirements reconstructed from repository evidence

The durable requirements artifact is `docs/product/database.md` (content first introduced as `docs/database.md` in `838db64`). Requirements evidenced there and in `backend/app/models/base.py` docstrings:

- **Multi-tenant isolation by denormalized** `company_id` on tenant business tables — isolation must not depend on join correctness.
- **UUID v7 primary keys** for time-ordered inserts.
- **Soft deletes** on business tables; **append-only** history for audit/activities/AI messages/notifications.
- **VARCHAR check-style enums** (`native_enum=False`) instead of native Postgres ENUMs.
- **Money as** `NUMERIC(14,2)` — never float.
- **Deterministic constraint naming** for reviewable Alembic diffs.
- Explicit **ON DELETE** policy matrix (CASCADE / RESTRICT / SET NULL by relationship class).
- **Per-tenant uniqueness** rules (email, SKU, invoice number, roles, etc.).
- **RBAC**: global permission catalog + per-company roles + user_roles.
- **Repository-level fail-closed** tenant filtering as the application choke point.

Alternatives evidenced by chosen standards

Documented choices imply rejected alternatives:

- Relying on parent joins for tenant filtering (rejected — denormalized `company_id`)
- Native Postgres ENUMs (rejected — VARCHAR check-style)
- Hard-deleting business data by default (rejected — soft delete)
- Mutating history tables like business rows (rejected — append-only)

Repository evidence

- `docs/product/database.md`
- `backend/app/models/base.py` (standards docstring)
- Commit `838db64`

Engineering review / remediation

Not applicable to this phase beyond later validation that implementation matches these requirements (see WAL-11/WAL-13 reports).

Validation

Requirements are validated by subsequent architecture/modeling/implementation matching `database.md`, and by isolation tests encoding tenancy requirements.

Outcome

A stable requirements baseline for the multi-tenant data platform.

Evidence limitations (shared)

- Part 03.0 code primarily landed in git commit `838db64` (2026-07-09) under the commit message “chore: validate and stabilize backend foundation”, bundled with broader foundation scaffolding. There is no separate contemporaneous commit uniquely titled “Part 03.0”.
- Linear issues WAL-7…WAL-17 were created on 2026-07-18 as a historical execution hierarchy after the schema already existed in the repository.
- No contemporaneous Part-03.0 independent-review finding IDs, meeting notes, or signed review PDF exist in the repository (unlike Part 03.1’s report pack).
- Structural integrity remediations (composite cross-tenant FKs, soft-delete-safe unique indexes, missing `created_by`/`updated_by` FKs) are **Part 03.1** evidence (`f521d38`, revisions `3a2aec52f6d2` / `217e90d00cc0`) and are not attributed as silent Part 03.0 remediations.

Phase-specific limitation

No separate contemporaneous “requirements sign-off” document exists beyond `database.md` and Linear WAL-8 closeout.